COS 433 Problem Set 4

Due date: Wednesday, 11/11/2026 at 11:59pm. See the course website for our submission and grading policies.

Proof citations. Your proofs may assume without proof any fact stated in lecture or stated on the problem sets. All other statements – including results from textbooks or other sources that were not mentioned in lecture – should be proved in your problem set write-up.

Grading Policy. A total score of 100 points constitutes full credit for the problem set. You are not expected to complete all of the problems. Feel free to work on whichever problems you prefer!

Problem 1: Homomorphic Encryption (20 points)

Homomorphic encryption schemes are encryption schemes that admit additional algebraic structure. For this problem, let us say that a public key encryption scheme $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ is *multiplicatively homomorphic* if for all valid $\mathsf{pk}, m_1, m_2 \in \mathcal{M}$ and r_1, r_2, there exists an r' such that

$$\mathsf{Enc}(\mathsf{pk}, m_1; r_1) \cdot \mathsf{Enc}(\mathsf{pk}, m_2; r_2) = \mathsf{Enc}(\mathsf{pk}, m_1 \cdot m_2; r')$$

(a) Show that both the Goldwasser-Micali encryption $(\mathsf{Gen}_{\mathsf{GM}}, \mathsf{Enc}_{\mathsf{GM}}, \mathsf{Dec}_{\mathsf{GM}})$ (for message space $\{\pm 1\}$) and the ElGamal encryption $(\mathsf{Gen}_{\mathsf{EG}}, \mathsf{Enc}_{\mathsf{EG}}, \mathsf{Dec}_{\mathsf{EG}})$ (for message space $\mathbb{QR}_p$) as seen in class are multiplicatively homomorphic.

(b) Prove that any multiplicatively homomorphic encryption scheme is *not* CCA-secure.

For reference, we say that a public key encryption scheme $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ is CCA-secure if for any polynomial time adversary $\mathcal{A}$, there exists a negligible function $\epsilon(\lambda)$ such that:

$$\Pr[G_{\mathcal{A}}(1^\lambda) = 1] \leq \frac{1}{2} + \epsilon(\lambda),$$

where $G_{\mathcal{A}}(1^\lambda)$ is the following game:

- Sample $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{Gen}(1^\lambda)$
- Run $(m_1, m_2) \leftarrow \mathcal{A}^{\mathsf{Dec}(\mathsf{sk}, \cdot)}(\mathsf{pk})$, where the adversary is given access to the decryption oracle $\mathsf{Dec}(\mathsf{sk}, \cdot)$ and input pk.
- Sample $b \leftarrow \{0, 1\}$ uniformly.

- Run $b^* \leftarrow \mathcal{A}^{\mathsf{Dec}(\mathsf{sk},\cdot)}(\mathsf{pk}, \mathsf{ct}^* = \mathsf{Enc}(\mathsf{pk}, m_b; r))$, where the adversary is given access to the decryption oracle $\mathsf{Dec}(\mathsf{sk},\cdot)$ and input pk and ct^*. Denote by Q the set of queries that the adversary makes to the decryption oracle.
- The game $G_{\mathcal{A}}(1^\lambda)$ outputs 1 if and only if $b = b^*$ and $\mathsf{ct}^* \notin Q$.

This definition is analogous to the definition of CCA-security for secret-key encryption.

Problem 2: RSA and Factoring (30 points)

Let $N = pq$ be a product of two distinct primes such that $p \equiv q \equiv 3 \pmod 4$. Then, we know that $|\mathbb{Z}_N^\times| = \phi(N) = (p-1)(q-1)$.

(a) Let $a \leftarrow \mathbb{Z}_N^\times$ be a uniformly random unit modulo N. Prove that the distribution of $a^{\phi(N)/4}$ modulo N is uniform over four distinct values.
Hint: use the Chinese remainder theorem.

(b) Describe a randomized polynomial-time algorithm with the following property: given input (N, x) such that x is an integer multiple of $\phi(N)$, the algorithm outputs the factorization (p, q) with high probability.

Hint: first solve the case where $x = \phi(N)$.

(c) Assume that factoring is hard, as in class. Prove that for any polynomial-time adversary $A(N, e)$ given an RSA public key $(N, e) \leftarrow \mathsf{Gen}(1^\lambda)$ as input, the probability that A outputs the secret key d is $\mathsf{negl}(\lambda)$.

Problem 3: Hard-core bit for discrete log (20 points)

For some prime $p < 2^\lambda$ and generator g for $\mathbb{Z}_p^\times$, define the function $f : \mathbb{Z}_{p-1} \to \mathbb{Z}_p^\times$ as $f(x) = g^x \bmod p$ and the "most significant bit" of x as

$$b(x) = \begin{cases} 0 & x < \frac{p-1}{2}, \\ 1 & x \geq \frac{p-1}{2}. \end{cases}$$

(a) Prove that if there exists a polynomial-time algorithm that always computes $b(x)$ correctly given $f(x)$ (along with (p, g)), then there is another polynomial-time algorithm that always inverts $f(x)$. **(20 points)**

Hint: Given an algorithm computing $b(x)$ from $g^x \pmod p$, describe an algorithm that on input g^x outputs $\mathsf{lsb}(x), g^{\lfloor x/2 \rfloor}$, where $\mathsf{lsb}(x)$ denotes the least significant bit of x.

(b) **Bonus:** Prove that $b(x)$ is a hard-core bit of $f(x)$ assuming $f(x)$ is one-way. In other words, show that if there exists a polynomial-time algorithm that computes $b(x)$ with probability $\frac{1}{2} + \epsilon(\lambda)$ for some non-negligible $\epsilon(\lambda)$ given $f(x)$, then there exists another polynomial-time algorithm that inverts $f(x)$ with non-negligible probability.

Problem 4: Lossy Encryption (30 points)

We say that a public-key encryption scheme $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ is *lossy* if there is a *fake key generation algorithm* $\mathsf{Gen}^*(1^\lambda) \to \mathsf{pk}^*$ that outputs a public key pk^*, *but no secret key*, satisfying the following two properties:

- The distribution of $\mathsf{pk}^* \leftarrow \mathsf{Gen}^*(1^\lambda)$ is computationally indistinguishable from that of a public key pk for $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$.
- When encrypting using $\mathsf{pk}^* \leftarrow \mathsf{Gen}^*(1^\lambda)$, encryptions of 0 and 1 are *statistically indistinguishable*. That is, the distributions

  $$\{\mathsf{pk}^* \leftarrow \mathsf{Gen}^*(1^\lambda) : (\mathsf{pk}^*, \mathsf{Enc}(\mathsf{pk}^*, 0))\} \approx_s \{\mathsf{pk}^* \leftarrow \mathsf{Gen}(1^\lambda) : (\mathsf{pk}^*, \mathsf{Enc}(\mathsf{pk}^*, 1))\}$$

  are (up to negligible advantage) indistinguishable to all (even inefficient) distinguishers.

(a) Show that any lossy encryption scheme satisfies IND-security (the basic PKE security property).

(b) Prove that a variant of the Goldwasser-Micali PKE scheme from class is a lossy encryption scheme under the quadratic residuosity assumption.

Hint: sample key pairs in a way that the QR assumption tells you something useful about the distribution of public keys.

(c) Construct a lossy encryption scheme assuming that the DDH assumption holds on some effective group $\mathbb{G}$ with known *prime* order ($|G_\lambda| = p_\lambda$ for every λ).

Hint: consider the matrix $\begin{bmatrix} 1 & x \\ y & z \end{bmatrix} \in \mathbb{Z}_p^{2\times 2}$ (for either $z = xy$ or uniformly random z). What happens if you right multiply this matrix by a random vector $\begin{bmatrix} r \\ s \end{bmatrix}$?

Problem 5: Bootstrapping One-Time Signature Schemes (50 points)

In this problem, we will explore how to "bootstrap" stronger digital signature schemes from weaker ones, such as the scheme we construct from CRHFs in class.

(a) Let $(\mathsf{Gen}, \mathsf{Sign}, \mathsf{Verify})$ denote a one-time digital signature scheme supporting messages of length $n(\lambda)$, such that for $(\mathsf{sk}, \mathsf{vk}) \leftarrow \mathsf{Gen}(1^\lambda)$, we have $|\mathsf{vk}| = m(\lambda)$.

Given (only) this scheme, construct a one-time digital signature scheme supporting messages of length $2 \cdot n(\lambda)$ such that $|\mathsf{vk}| = 2 \cdot m(\lambda)$. Prove that your scheme is secure. **(10 points)**

(b) Suppose that $(\mathsf{Gen}, \mathsf{Sign}, \mathsf{Verify})$ is actually many-time secure, satisfying existential unforgeability. Is the scheme you constructed in part (a) many-time secure? Prove or disprove. **(10 points)**

(c) Now, suppose that $n(\lambda) \geq 2 \cdot m(\lambda)$. Given the original $(\mathsf{Gen}, \mathsf{Sign}, \mathsf{Verify})$ as in part (a), construct a one-time digital signature scheme supporting messages of length $2 \cdot n(\lambda)$ such that $|\mathsf{vk}| = m(\lambda)$. Prove that your scheme is secure. **(10 points)**

Hint: make use of randomness in your signing algorithm.

(d) Under the same hypotheses as part (c), construct a one-time digital signature scheme supporting messages of arbitrary $\mathsf{poly}(\lambda)$ length, such that $|\mathsf{vk}| = m(\lambda)$. We will give full credit for a secure construction without proof, and give bonus points for a security proof. **(10 points)**

(e) Under the same hypotheses as part (c), construct a full-fledged (*many-time* secure) digital signature scheme with messages of arbitrary polynomial length. You may additionally assume the existence of a PRF. **(10 points)**

Combined with the construction from lecture, this gives a construction of digital signatures from any CHRF family.

Problem 6: CRHFs and OWFs (20 points)

Let $\mathcal{H} = \{\{H_k : \{0,1\}^{2\lambda} \to \{0,1\}^{\lambda}\}_{k \in \{0,1\}^{\lambda}}\}_{\lambda}$ be a family of collision-resistant hash functions with uniformly random keys $k \leftarrow \{0,1\}^{\lambda}$. Prove that the function

$$f(k, x) = (k, H_k(x))$$

is a one-way function.